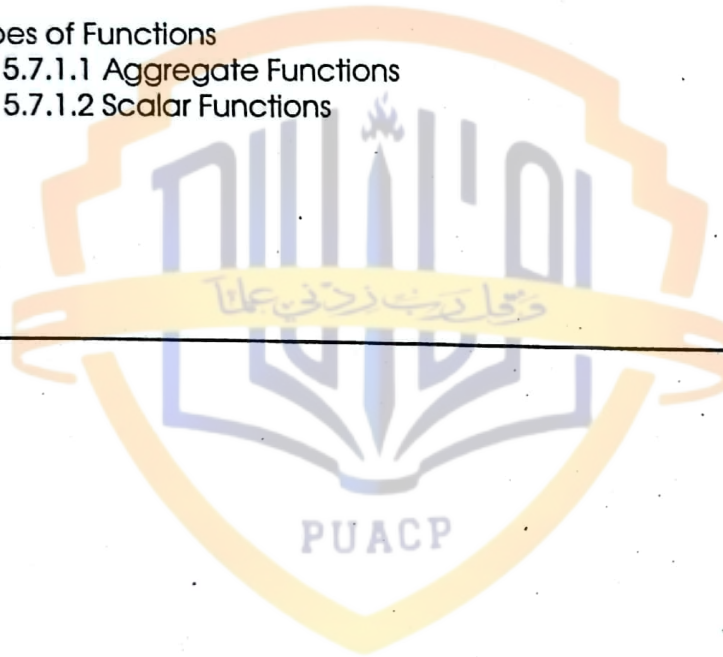


Chapter Overview

- 15.1 Structured Query Language
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- 15.7 Functions
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Short Questions
Multiple Choices
Fill in the Blanks
True/False



15.1 Structured Query Language

SQL stands for Structured Query Language. SQL is a standard computer language for accessing and manipulating database systems. SQL statements are used to retrieve and update data in a database. SQL works with database programs like MS Access, DB2, Informix, MS SQL Server, Oracle, Sybase, etc. SQL statements are called queries. These statements are used to retrieve, update, insert and delete records from tables in database.

- **SELECT** - extracts data from a database table
- **UPDATE** - updates data in a database table
- **DELETE** - deletes data from a database table
- **INSERT INTO** - inserts new data into a database table

15.2 The SELECT Statement

SELECT statement is used to select data from a table.

Syntax

SELECT column_name(s) FROM table_name;

Select Some Columns

To select the columns named "LastName" and "FirstName", use a SELECT statement like this:

SELECT LastName, FirstName FROM Authors;

"Authors" Table

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer
Mustafa	Tasleem	Gulistan Colony	Faisalabad	Assistant Professor
Raza	Ahsan	Peoples Colony	Faisalabad	Lecturer

Result

LastName	FirstName
Mahmood	Tariq
Saeed	Imran
Mustafa	Tasleem
Raza	Ahsan

Select All Columns

A * symbol is used to select all columns from a table as follows:

SELECT * FROM Authors;

Result

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer
Mustafa	Tasleem	Gulistan Colony	Faisalabad	Assistant Professor
Raza	Ahsan	Peoples Colony	Faisalabad	Lecturer

15.3 The WHERE Clause

WHERE clause is used to select data from a table conditionally.

Syntax

SELECT column FROM table WHERE condition;

The following operators can be used with the WHERE clause:

Operator	Description
=	Equal
<>	Not equal
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal
BETWEEN	Between a range
LIKE	Search for a pattern

Example 15.1

WHERE clause is added to SELECT statement to select only those authors whose designation is "lecturer":

SELECT * FROM Authors WHERE Designation='Lecturer';

Result

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer
Raza	Ahsan	Peoples Colony	Faisalabad	Lecturer

Using Quotes

SQL uses single quotes around text values. Most database systems will also accept double quotes. Numeric values should not be enclosed in quotes.

The LIKE Condition

The LIKE condition is used to specify a search for a pattern in a column.

Syntax

SELECT column FROM table WHERE column LIKE pattern;

A "%" sign can be used to define wildcards both before and after the pattern.

Example 15.2

The following SQL statement will return authors with first names that start with an 'O':

SELECT * FROM Authors WHERE FirstName LIKE 'O%';

The following SQL statement will return authors with first names that end with an 'a':

SELECT * FROM Authors WHERE FirstName LIKE '%a'

The following statement will return authors whose first names contain the pattern 'ri':

SELECT * FROM Authors WHERE FirstName LIKE '%ri%'

15.4 The INSERT INTO Statement

The INSERT INTO statement is used to insert new rows into a table.

Syntax

INSERT INTO table_name VALUES (value1, value2,...)

You can also specify the columns for which you want to insert data:

INSERT INTO table_name (column1, column2,...) VALUES (value1, value2,...)

Example 15.3

This SQL statement:

INSERT INTO Authors VALUES ('Bayross', 'Ivan', 'G-11', 'London', 'Professor')

will add a new record in the table:

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer
Mustafa	Tasleem	Gulistan Colony	Faisalabad	Assistant Professor
Raza	Ahsan	Peoples Colony	Faisalabad	Lecturer
Bayross	Ivan	G-11	London	Professor

Insert Data in Specified Columns

You can also insert values only in the specified fields. For example, if you want add only FirstName and Designation in a new record then you have to mention the column names in the Insert statement.

Example 15.4

This SQL statement:

INSERT INTO Authors (LastName, Designation) VALUES ('Deitel', 'Professor')

will give this result:

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer
Mustafa	Tasleem	Gulistan Colony	Faisalabad	Assistant Professor
Raza	Ahsan	Peoples Colony	Faisalabad	Lecturer
Bayross	Ivan	G-11	London	Professor
	Deitel			Professor

15.5 The Update Statement

The UPDATE statement is used to modify the data in a table.

Syntax

UPDATE table_name SET column_name = new_value
WHERE column_name = some_value;

Update one Column in a Row

We want to modify the last name of the person with first name of "Ahsan":

```
UPDATE Authors SET LastName = 'Sattar'
WHERE FirstName = 'Ahsan';
```

Result

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer
Mustafa	Tasleem	Gulistan Colony	Faisalabad	Assistant Professor
Sattar	Ahsan	Peoples Colony	Faisalabad	Lecturer
Bayross	Ivan	G-11	London	Professor
	Deitel			Professor

Update several Columns in a Row

We want to change the address and add the name of the city:

```
UPDATE Authors SET Address = 'Zero Point', City = 'New York'
WHERE LastName = 'Bayross'
```

Result

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer
Mustafa	Tasleem	Gulistan Colony	Faisalabad	Assistant Professor
Raza	Ahsan	Peoples Colony	Faisalabad	Lecturer
Bayross	Ivan	Zero Point	New York	Professor
	Deitel			Professor

15.6 The Delete Statement

The DELETE statement is used to delete rows in a table.

Syntax

```
DELETE FROM table_name WHERE column_name = some_value
```

Example 15.5

"Deitel" will be deleted with the following statement:

```
DELETE FROM Authors WHERE FirstName = 'Deitel';
```

Delete All Rows

It is possible to delete all rows in a table without deleting the table. This means that all data in the table will be deleted but the table structure, attributes, and indexes will be intact:

```
DELETE FROM table_name; OR DELETE * FROM table_name;
```

Sort the Rows

The ORDER BY clause is used to sort the rows.

Example 15.6

The following statement will display the authors in alphabetical order with respect to first name:

```
SELECT * FROM Authors
ORDER BY FirstName;
```

Result

LastName	FirstName	Address	City	Designation
Raza	Ahsan	Peoples Colony	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer
Bayross	Ivan	Zero Point	New York	Professor
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Mustafa	Tasleem	Gulistan Colony	Faisalabad	Assistant Professor

Example 15.7

You can give more than one column for sorting. The preference is given to first column. If values in the first column are similar then the records will be sorted with respect to the second column. The following statement will display the table in descending order with respect to the Designation. If designation is similar then records will be sorted by LastName.

```
SELECT * FROM Authors ORDER BY Designation, LastName DESC;
```

Result

LastName	FirstName	Address	City	Designation
Bayross	Ivan	Zero Point	New York	Professor
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer
Raza	Ahsan	Peoples Colony	Faisalabad	Lecturer
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Mustafa	Tasleem	Gulistan Colony	Faisalabad	Assistant Professor

AND & OR

AND and OR are used to give two or more conditions in a WHERE clause. The AND operator displays a row if ALL conditions listed are true. The OR operator displays a row if ANY of the conditions listed are true.

Example 15.8

The following query uses AND to display the author with the first name equal to "Tariq", and the last name equal to "Mahmood":

```
SELECT * FROM Authors
WHERE FirstName='Tariq'
AND LastName='Mahmood';
```

Result

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer

Example 15.9

The following example uses OR to display the author with the first name equal to "Tariq", or the last name equal to "Saeed":

```
SELECT * FROM Authors
WHERE FirstName='Tariq'
OR LastName='Saeed';
```

Result

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer

Example 15.10

You can also combine AND and OR in one statement. You can use parentheses to form complex expressions:

```
SELECT * FROM Authors WHERE
(FirstName='Tariq' OR FirstName='Imran')
AND Designation = 'Lecturer';
```

Result

LastName	FirstName	Address	City	Designation
Mahmood	Tariq	Madina Town	Faisalabad	Lecturer
Saeed	Imran	Gulistan Colony	Faisalabad	Lecturer

BETWEEN...AND

The BETWEEN...AND operator selects a range of data between two values. These values can be numbers, text, or dates.

```
SELECT column_name FROM table_name
WHERE column_name
BETWEEN value1 AND value2
```

Students Table

Roll No	Name	Marks	Grade
1	Usman	100	A
2	Ali	55	C
3	Nadeem	72	B
4	Adnan	61	B
5	Waqar	90	A

Example 15.11

The following query will display the students with marks between 60 and 80:

```
SELECT * FROM Students WHERE Marks
BETWEEN 50 AND 80;
```


Result

Roll No	Name	Marks	Grade
2	Ali	55	C
3	Nadeem	72	B
4	Adnan	61	B

Example 15.12

The following example will display the students outside the range used in the previous example by using the NOT operator:

SELECT * FROM Students WHERE Marks NOT BETWEEN 50 AND 80;

Result

Roll No	Name	Marks	Grade
1	Usman	100	A
5	Waqar	90	A

15.7 Functions

There are different built-in functions that can be used to perform different tasks easily. The syntax for built-in SQL functions is:

SELECT function(column) FROM table;

15.7.1 Types of Functions

There are several basic types and categories of functions in SQL. The basic types of functions are:

- Aggregate Functions
- Scalar functions

15.7.1.1 Aggregate Functions

Aggregate functions operate against a collection of values, but return a single value.

Function	Description
AVG(column)	Returns the average value of a column
COUNT(column)	Returns the number of rows (without a NULL value) of a column
COUNT(*)	Returns the number of selected rows
LAST(column)	Returns the value of the last record in the specified field
MAX(column)	Returns the highest value of a column
MIN(column)	Returns the lowest value of a column
SUM(column)	Returns the total sum of a column

15.7.1.2 Scalar Functions

Scalar functions operate against a single value, and return a single value based on the input value.

Useful Scalar Functions

Function	Description
UCASE(c)	Converts a field to upper case
LCASE(c)	Converts a field to lower case
MID(c,start[,end])	Extract characters from a text field
LEN(c)	Returns the length of a text field
INSTR(c)	Returns the numeric position of a named character within a text field
LEFT(c,number_of_char)	Return the left part of a text field requested
RIGHT(c,number_of_char)	Return the right part of a text field requested
ROUND(c,decimals)	Rounds a numeric field to the number of decimals specified
MOD(x,y)	Returns the remainder of a division operation
NOW()	Returns the current system date

Example 15.13

The following example will get the sum of all students:

```
SELECT SUM(Marks) FROM Students;
```

The following example will get the marks of the student with maximum marks:

```
SELECT MAX(Marks) FROM Students;
```

The following example will get the average of all marks of the students:

```
SELECT AVG(Marks) FROM Students;
```

Short Questions

Q.1. What is SQL?

SQL stands for Structured Query Language. It is used to manipulate databases. It was developed by IBM in 1970s.

Q.2. List out some uses of SQL.

SQL is used to create and modify database structure, set security settings, add user permission on databases or tables, query a database for information and add, delete and update the contents of a database.

Q.3. List out different ways of interacting with database.

There are two ways of interacting with databases. These are using Integrated Development Environment (IDE) and using Data Access Language.

Q.5. Which command is used to create a table in SQL?

CREATE TABLE statement is used to create new table in a database.

Q.6. Write the syntax of CREATE TABLE command.

The syntax of this statement is as follows:

```
CREATE TABLE Table_Name
```

```
(field_1 data_type, field_2 data_type, ... field_N data_type);
```


Q.7. Give an example of CREATE TABLE command.

```
CREATE TABLE Student
```

```
(roll_no INT(5), name CHAR(15), date_of_birth DATE, phone_no CHAR(12));
```

Q.8. How is primary key specified in SQL?

Primary key is specified in CREATE TABLE statement. The keyword PRIMARY KEY is used for this purpose.

Q.9. Give an example to specify primary key in CREATE TABLE command.

```
CREATE TABLE Student
```

```
(roll_no INT(5) PRIMARY KEY, name CHAR(15), date_of_birth DATE);
```

Q.13. What is NUMBER data type of SQL?

NUMBER data type is used to store numeric value. The size indicates the maximum number of digits.

Q.14. What is DATE data type of SQL?

DATE data type is used to store date values.

Q.15. How is data added in tables using SQL?

Data can be added in tables by using INSERT INTO command of SQL. This statement gets data from the user as parameters. It stores the data in the table as a new record.

Q.16. What is the syntax of INSERT INTO command?

The syntax of INSERT INTO statement is as follows:

```
INSERT INTO Table_Name
```

```
VALUES (value1, value2, ..., valueN)
```

Q.17. Give an example of INSERT INTO command.

```
INSERT INTO Student
```

```
VALUES (1, 'Usman Khalil', 'Muhammad Khalil', '21/05/1996', '733474');
```

Q.18. How is data retried from tables using SQL?

The data can be retrieved from tables by using SELECT statement. This statement retrieves data from tables and displays it to the user.

Q.19. What is the syntax of SELECT statement?

The syntax of SELECT statement is as follows:

```
SELECT Field_Name FROM Table_Name;
```

Q.20. Give an example of SELECT statement.

The following example retrieves roll_no and name from Student table:

```
SELECT roll_no, name FROM Student;
```

Q.21. What is the use of * in SELECT statement?

The asterisk "*" is used in place of Field_Name to retrieve all fields of a table. The following example will retrieve all fields of Student table.

```
SELECT * FROM Student;
```

Q.22. How is condition used in SELECT statement?

WHERE clause is used in SELECT statement to retrieve data that matches with a condition.

Q.23. Write the syntax of WHERE clause in SELECT statement.

The syntax of using WHERE clause is as follows:

```
SELECT Field_Name FROM Table_Name WHERE condition;
```


Q.24. Give an example of using WHERE clause.

The following example retrieves only those students whose roll_no are greater than 3:

SELECT * FROM Student WHERE roll_no > 3;

Q.25. How is data updated in tables using SQL?

The data can be updated in tables by using UPDATE statement. This statement is used to change the values of fields in the existing records.

Q.26. Write the syntax of UPDATE statement.

The syntax of UPDATE statement is as follows:

UPDATE Table_Name SET Field_Name=New_Value WHERE condition;

Q.27. Give an example of UPDATE statement.

The following example updates the marks of the student with roll_no 5:

UPDATE Student SET Marks = 100 WHERE roll_no = 5;

Q.28. What happens if WHERE clause is not used in UPDATE statement.

If WHERE clause is not used in UPDATE statement, it updates all records in the table.

Q.29. How is data deleted in tables using SQL?

The data can be deleted in tables by using DELETE statement. This statement is used to delete the existing records.

Q.30. What is the syntax of DELETE statement?

The syntax of DELETE statement is as follows:

DELETE FROM Table_Name WHERE condition;

Q.31. Give an example of DELETE statement.

The following example deletes the record of the student with roll_no 5:

Multiple Choice

- SQL is:
 - Procedural language
 - Non-procedural language
 - Not a data sub language
 - None
- Which SQL keyword is used to define new table and describe table's columns?
 - Set
 - Create
 - Alter
 - Modify
- Which SQL keyword is used to delete a table's structure?
 - Delete
 - Drop
 - Update
 - Alter
- In a SQL query, which keyword is used to specify the table(s) to be used?
 - EXISTS
 - Update
 - From
 - Set
- The asterisk (*) means that:
 - All columns of table are required
 - Records meeting the criteria are required
 - Records meeting partial criteria are required
 - None of the above
- Which SQL keyword is used to specify condition in query?
 - EXISTS
 - Where
 - From
 - Select
- Which SQL keyword is used to add one or more rows to a table?
 - MODIFY
 - INSERT
 - SELECT
 - INSERT
- Which SQL keyword is used to change one or more rows in a table?
 - MODIFY
 - INSERT
 - SELECT
 - UPDATE

9. Which SQL keyword is used to change a column value?
a. CHANGE b. INSERT c. SELECT d. SET
10. Which keyword is used to remove one or more rows from a table?
a. CHANGE b. Delete c. SELECT d. Erase
11. Which statement will retrieve EmployeeID, EmployeeName and SSN from Employees?
a. Select Employees.EmployeeID, Employees.EmployeeName, Employees.SSN
b. Select EmployeeID, EmployeeName, SSN
c. Select EmployeeID, EmployeeName, SSN from Employees
d. Select Employees: EmployeeID, EmployeeName, SSN
12. The correct syntax to retrieve all fields from a table named Employees for all employees with a last name of Abdullah is:
a. Select * from Employees Where LastName = 'Abdullah'
b. Select All from Employees Where LastName = Abdullah
c. Select * from Employees Where LastName is 'Abdullah'
d. Select * from Employees Order By LastName = 'Imran'

Answers

1. b	2. b	3. b	4. c
5. a	6. b	7. d	8. d
9. d	10. b	11.	12.

Fill in the Blanks

1. SQL stands for _____.
2. SQL was developed by _____ corporation in 1970.
3. Sybase & SQL Server are different types of _____ programs.
4. SQL is a _____ oriented database language used to perform DBMS operations.
5. Generally, there are _____ ways of interacting with the databases.
6. IDE stands for _____.
7. IDE is a _____ facility used to communicate with the database.
8. _____ language is used to manipulate data in a database.
9. SQL interface takes command from user and converts it into _____ database command.
10. _____ statement is used to create a new table in database using SQL.
11. _____ statement of SQL contains name of the table, its fields & data types to be created.
12. To create a primary key in CREATE TABLE statement of SQL, _____ keyword is used.
13. To store a variable length character string in table _____ data type is used.
14. Data can be added in tables by using _____ command of SQL.
15. To retrieve data from tables, _____ statement of SQL is used.
16. To retrieve all fields of a table using SQL, _____ is used in place of field name.
17. FROM clause is used in _____ statement to retrieve data from a table.
18. _____ clause is used in SELECT statement to retrieve data that matches a particular condition.

Answers

1. Structured Query Language	2. IBM	3. Database
4. Set	5. Two	6. Integrated Development Environment
7. Graphical	8. Data Access	9. Internal
10. CREATE TABLE	11. CREATE TABLE	12. PRIMARY KEY
13. VARCHAR()	14. INSERT INTO	15. SELECT
16. Asterisk	17. SELECT	18. WHERE

True / False

1. The most important query language is Oracle.
2. SQL is a procedural language.
3. SQL stands for Standard Query Language.
4. SQL is a data sublanguage or data access language.
5. The result of every SQL query is a relation.
6. SQL was developed by INTEL Corporation.
7. Sybase and SQL Server are different types of graphic programs.
8. IDE is a graphical facility used to communicate with the database.
9. Data Access language is used to manipulate data in a database.
10. CREATE TABLE statement of SQL is used to create a new table in database.
11. To store a fixed length character string in a table FCHAR data type is used in CREATE TABLE statement of SQL.
12. INSERT FROM command of SQL is used to add data in table.

Answers

1. F	2. F	3. F	4. F
5. T	6. F	7. F	8. T
9. T	10. T	11. F	12. F